

Management of Type 2 Diabetes

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Block: Endocrine/Urologic/Reproductive

Thread: Pharmacology

Conflict of interest: Dr. Simon has no conflicts of interest to disclose.

Learning Objectives

By exam time, you should be able to do the following:

1. Explain the mechanism of action of the non-insulin therapies for diabetes
2. Describe the clinical use and efficacy of each class of drugs
3. Identify the most common side effects of each drug class and use adverse-effect profiles to guide drug selection in clinical scenarios.
4. Identify the contraindications for each drug class
5. Formulate a therapeutic management plan to treat type 2 diabetes

Further Reading

1. [Masharani U, Kroon L. Pancreatic Hormones & Glucose-Lowering Drugs. In: Vanderah TW. eds. *Katzung's Basic & Clinical Pharmacology*, 16th Edition. McGraw-Hill; 2024. Accessed September 29, 2026.](#)
2. [American Diabetes Association. *Standards of care in diabetes—2026. Diabetes Care*. 2026;49\(suppl 1\).](#)

Study Questions

1. You are starting a patient on metformin. They want to know what the most common side effect is. You tell them:
 - a. Gastrointestinal side effects
 - b. Lactic acidosis
 - c. Kidney dysfunction
 - d. Hypoglycemia
2. A patient is started on an SGLT-2 inhibitor. They want to know the effects it will have on their metabolic profile. You tell them:
 - a. Decreased a1c, decreased weight, decreased blood pressure
 - b. Decreased a1c, increased weight, decreased blood pressure
 - c. Decreased a1c, decreased weight, increased blood pressure
 - d. Decreased a1c, increased weight, increased blood pressure
3. A patient with type 2 diabetes asks you to provide them with a medication regimen that will help them to lose weight. The following combination of medications is most likely to accomplish that goal:
 - a. Metformin, GLP-1 analog, SGLT-2 inhibitor
 - b. Metformin, sulfonylurea, bromocriptine
 - c. Metformin, thiazolidinedione, colesvelam
 - d. Metformin, repaglinide, pramlintide
4. You are treating a 74 - year old type 2 diabetic patient with metformin and glipizide 10mg once daily. Their fasting sugars are 90-130 mg/dL and their a1c is 6.5%. However, they get frequent hypoglycemia before lunch. The best advice to give the patient is:
 - a. Hold the metformin
 - b. Decrease the dose of metformin
 - c. Eat a mid-morning snack
 - d. Hold the glipizide

5. You start a 35 - year old patient with type 2 diabetes on a GLP-1 analog. The medical student rotating on your service asks you to explain the metabolic effects of this class of medications. You tell them that they:
- a. Increase insulin and glucagon secretion in a glucose dependent fashion.
 - b. Increase insulin secretion and decrease glucagon in a glucose dependent fashion.
 - c. They are insulin analogs that acts like insulin in the body.
 - d. Increase insulin secretion and decrease glucagon in a glucose independent fashion.

A. Why Know Diabetes?

- a. Prevalence: 38.4 million people have diabetes (11.6% of the US population), and 8.7 million adults who are undiagnosed
- b. Diabetes is the 8th leading cause of death
- c. Heart disease and stroke are 2x more likely in diabetics
- d. Diabetes is the Leading cause of: **End Stage Renal Disease**, new cases of **blindness** in adults, and of non-traumatic lower limb **amputations**
- e. \$413 Billion = total direct estimated costs of diagnosed diabetes (in 2017)

If Blood Sugars are controlled, this dramatically reduces the risk for microvascular complications in both Type 1 and Type 2 Diabetes as shown in two landmark studies:

- a. The Diabetes Control and Complications Trial (DCCT) studied Type 1 diabetics. Comparing patients who intensively managed their glucose levels (A1c of 7.2%) to those who did usual management (A1c of 9.2%), there was a lower incidence of retinopathy and microalbuminuria, reduced progression of existing retinopathy and nephropathy, and a reduced incidence of neuropathy in the intensive management group.
- b. The United Kingdom Prospective Diabetes Study (UKPDS) was a similar study in Type 2 patients, and found a 25 % risk reduction in microvascular disease in the intensively managed group. A 1 percent fall in A1c was associated with 35% reduction in microvascular endpoints

B. Prevention of Diabetes

- a. Great evidence for screening. Weight loss and lifestyle changes are effective preventive strategies
 - i. Testing for prediabetes and/or type 2 diabetes in asymptomatic people should be considered in adults of any age with overweight or

obesity (BMI ≥ 25 kg/m² or ≥ 23 kg/m² in Asian Americans) who have one or more risk factors

- ii. For all adults, screening should start at age 35 (per the ADA) or age 40 (USPSTF)

b. The Diabetes Prevention Program

- i. Patients with prediabetes should be referred to an intensive behavioral lifestyle intervention program to achieve and maintain 7% loss of initial body weight and increase moderate-intensity physical activity 150 min/ week (such as brisk walking)
- ii. Can reduce the risk by 58%!

C. Pathophysiologic Defects in Type 2 Diabetes

- a. It is important to review the defects that lead to glucose intolerance and diabetes, as this can help to understand the mechanisms of action and inform the selection of pharmacologic agents we use to treat Type 2 Diabetes.
- b. Insulin resistance in the muscle and liver and β -cell failure represent the core pathophysiologic defects in type 2 diabetes, other defects that also contribute to hyperglycemia are:
 - i. the fat cell (accelerated lipolysis)
 - ii. gastrointestinal tract (incretin deficiency/resistance)
 - iii. α -cell (hyperglucagonemia),
 - iv. kidney (increased glucose reabsorption)
 - v. brain (insulin resistance)

D. Type 2 Diabetes Management Strategies

- 1. Patients with T2DM are 2 times more likely to develop cardiovascular (CV) diseases due to increased risk factors (hypertension, obesity, hyperlipidemia)
 - a. Don't forget that these other risk factors should also be addressed (control of hypertension, statin use, weight loss, smoking cessation)

2. The American Diabetes Association “Use of Glucose-Lowering Medications in the Management of Type 2 Diabetes”
 - a. Healthy Lifestyle behaviors / SDOH (Social determinants of health) and Diabetes Education
 - b. Cardiorenal risk reduction in high risk patients
 - As we go through pharmaceutical classes- be mindful of EXTRAGLYCEMIC benefits
 - c. Glycemic Management
 - d. Minimize Weight Gain/ Promote Weight loss
 - e. Consider Cost and Access
 - f. Minimize Hypoglycemia

diabetesjournals.org/care

Pharmacologic Approaches to Glycemic Treatment S191

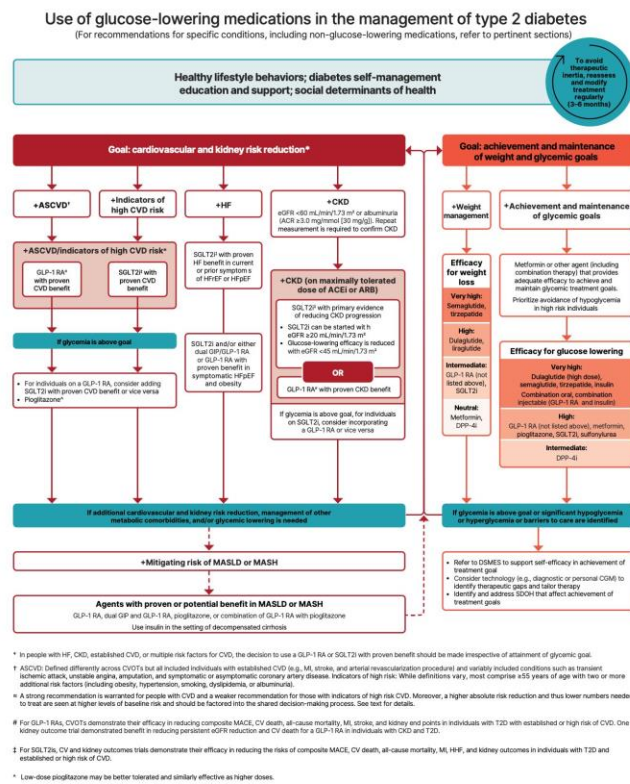


Figure 9.4—Use of glucose-lowering medications in the management of type 2 diabetes. The left side of the algorithm prioritizes mitigation of diabetes-related complications and end-organ effects, while the right side addresses weight and glucose management goals. ACEi, angiotensin-converting enzyme inhibitor; ACR, albumin-to-creatinine ratio; ARB, angiotensin receptor blocker; ASCVD, atherosclerotic cardiovascular disease; CSM, continuous glucose monitoring; CKD, chronic kidney disease; CV, cardiovascular; CVD, cardiovascular disease; CVOT, cardiovascular outcomes trial; DPP-4i, dipeptidyl peptidase 4 inhibitor; DSMEs, diabetes self-management education and support; eGFR, estimated glomerular filtration rate; GLP-1 RA, glucagon-like peptide 1 receptor agonist; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; HFrEF, hospitalization for heart failure; MACE, major adverse cardiovascular events; MASLD, metabolic dysfunction-associated steatotic liver disease; MASLD, metabolic dysfunction-associated steatotic liver disease; MI, myocardial infarction; SDOH, social determinants of health; SGLT2i, sodium-glucose cotransporter 2 inhibitor; T2D, type 2 diabetes. Adapted from Davies et al. (90).

E. Lifestyle Modification/ Weight Loss

1. **MNT** (Medical Nutrition Therapy) delivered by a registered dietitian is associated with A1C decreases of: 0.3–2%
 - a. A variety of eating patterns are acceptable (Mediterranean, Dietary Approaches to Stop Hypertension (DASH) , plant-based diets, portion control, reducing high glycemic index foods
2. **Physical activity**
 - a. 150 min or more of moderate-to-vigorous intensity aerobic activity per week (most adults)
 - b. Shorter durations (75min/ week) of vigorous-intensity/ interval training may be sufficient for younger and more physically fit individuals
 - c. Prolonged sitting should be interrupted every 30 min

F. Biguanides – Metformin

- a. **FIRST LINE AGENT** –we usually start metformin at diagnosis of Type 2 diabetes (ADA 2006)
- b. Weight neutral or modest weight loss
- c. Does not cause hypoglycemia
- d. Does not have adverse cardiovascular effects, appears to decrease cardiovascular events in certain populations
- e. Inexpensive!
 - i. **Mechanism of Action:** decrease hepatic glucose output/ inhibits gluconeogenesis. It may increase insulin-mediated glucose utilization in peripheral tissues
 - ii. **Clinical efficacy:** A1c lowering (approx. 1.5%)
 - iii. **Metabolism:** not metabolized and is excreted by the kidneys as the active compound.
 1. eGFR 45-60 mL/min per 1.73 m²- safe use
 2. eGFR 30-44- cautious use (max dose 1000mg/day)
 3. eGFR < 30 – contraindicated
 - a. Temporarily hold for radiocontrast administration

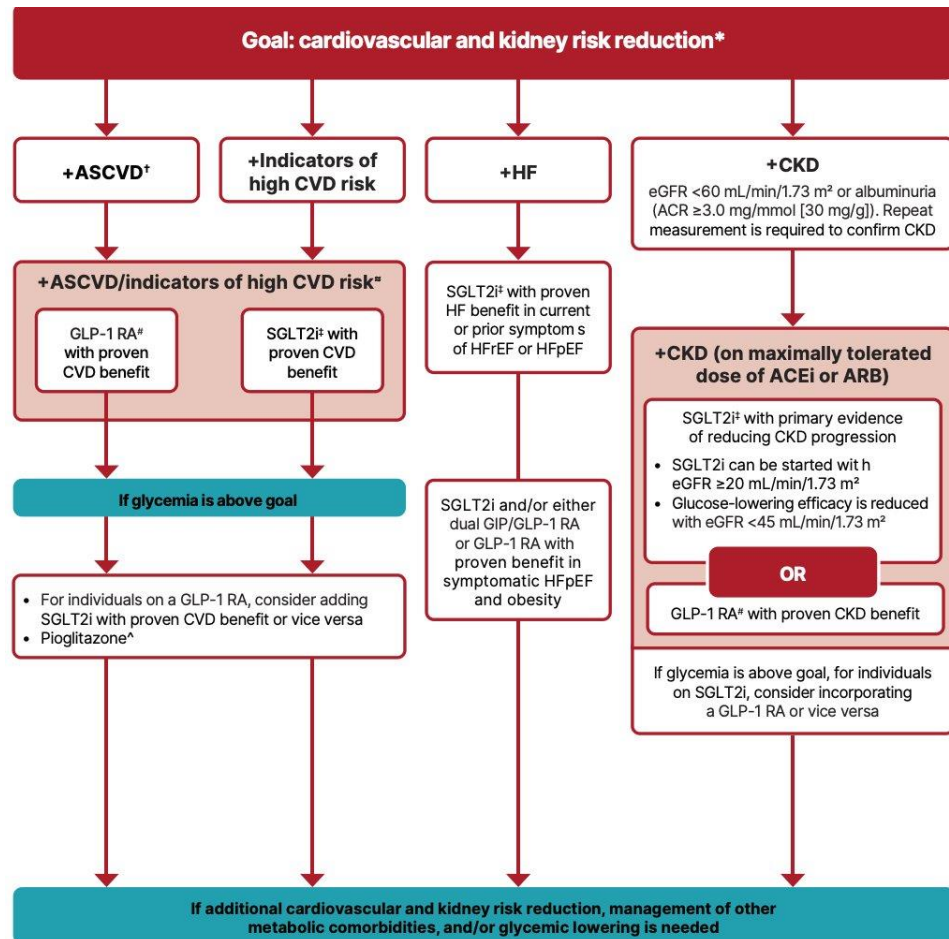
- b. Check renal function at least annually

f. Side Effects:

- i. **Gastrointestinal** Most common (up to 20%) diarrhea, abdominal discomfort, nausea (taking with food and slowly up titration of the dose can help alleviate)
- ii. **B12 deficiency** (after many years of use) –may interfere with absorption of B₁₂–intrinsic factor complex in the terminal ileum (periodic screening for B₁₂ deficiency should be considered, especially in patients with peripheral neuropathy)
- iii. **Lactic Acidosis** Rare but dangerous side effect. As a consequence of Metformin blocking gluconeogenesis, it may impair the hepatic metabolism of lactic acid
- iv. Almost all reported cases have involved patients with associated risk factors that should have contraindicated its use, such as conditions of tissue hypoxia when there is increased production of lactic acid (ex. heart failure, shock, sepsis, active liver disease, active alcohol abuse) and in renal failure when there is decreased clearance of metformin.

G. Selecting Therapies

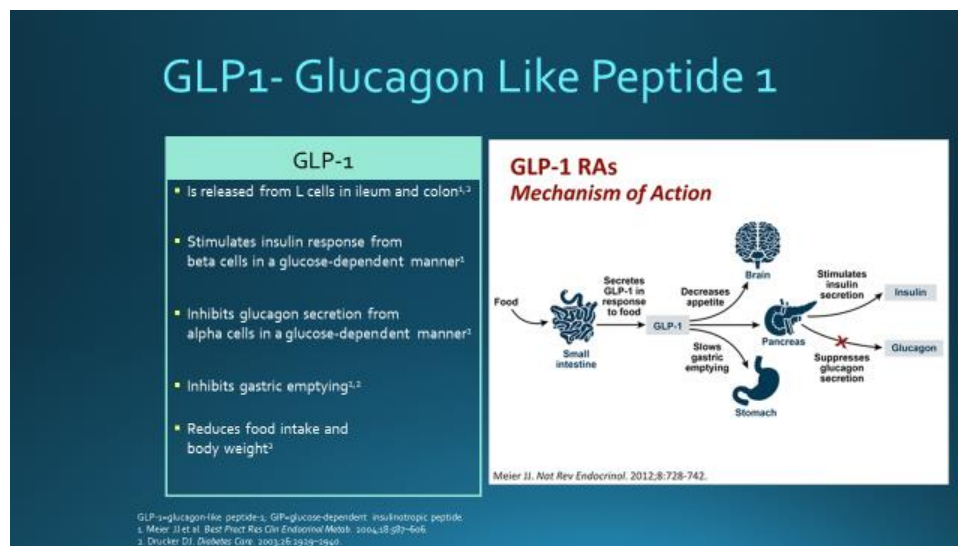
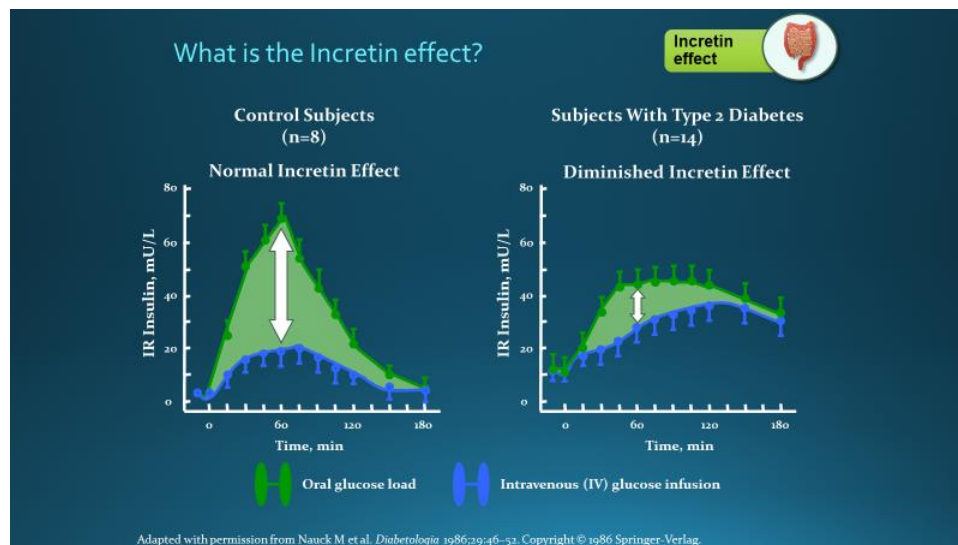
1. Prior guidelines advised metformin at diagnosis, and most often you will see that continuing
2. However, new guidelines reinforce that you also need to evaluate comorbidities, and this can guide you to specific medications
 - a. Does your patient have ASCVD, CHF, or CKD?
 - b. Does your patient have high risk for these?
 - i. ≥ 55 with 2 or more RF (obesity HTN, smoking, dyslipidemia, or albuminuria?)



H. Incretins

1. Gut Hormones which normally help with our glucose control
 - a. Stimulation of insulin secretion **only when blood sugar is high/ fed state** (glucose dependent)
 - b. Lowers both post meal and fasting glucose
 - c. Suppresses inappropriately elevated glucagon post-meal
 - d. Reductions in weight
2. Diabetics -demonstrated to exhibit almost total loss of incretin effect
 - a. GLP1: Glucagon Like Peptide 1
 - b. GIP: glucose-dependent insulinotropic peptide

3. The Incretin Effect



I. GLP-1 Receptor Agonists (GLP1RA) (“tides”)

1. May be most known for weight loss effects
2. Do not cause hypoglycemia
3. Next to insulin, the strongest for A1c lowering
4. Cardiovascular benefits in RCT’s
5. Can be expensive
 - a. **Mechanism of Action:** Synthetic glucagon-like peptide-1 (GLP-1) receptor agonists bind to the GLP-1 receptor and exhibit the effects of GLP1. They are resistant to degradation by the enzyme dipeptidyl peptidase-4 (DPP-4) and therefore have a longer half-life

- b. **Clinical Use:** all are injectable except oral semaglutide

Daily use

Exenatide

Liraglutide*

Semaglutide Oral*

Weekly Use

Exenatide LAR

Dulaglutide*

Semaglutide*

* **Cardiovascular benefits** – liraglutide, semaglutide and dulaglutide have been associated with improved cardiovascular outcomes; studies show they reduce MACE (Major Adverse Cardiac Events (death from cardiovascular cause, nonfatal MI, nonfatal stroke)

- c. **Glycemic efficacy** - In comparison with placebo, all GLP-1 receptor agonists reduced glycated hemoglobin (A1C) by approximately 1 percentage point or more
- d. **Weight loss** – Patients lose at least 2-3 kg in most trials, some trials / depending on the agent, weight loss is much greater
- e. **Side Effects/ Adverse Effects:**
- i. **Gastrointestinal** - nausea, vomiting, and diarrhea. Occur consistently in trials in 10 to 50% of patients (remember they slow gastric emptying)
 - ii. **Pancreas** – Acute pancreatitis has been reported *in association* with GLP-1 agonist treatment, warning against GLP-1 receptor agonist use in patients with a history of pancreatitis
 - iii. GLP-1 agonists are contraindicated in patients with a personal or family history of **medullary thyroid cancer** or multiple endocrine neoplasia 2A or 2B. In rodent studies hyperplasia of thyroid C cells was noted, and therefore a potential for the risk of medullary thyroid cancer

J. Selecting Therapies

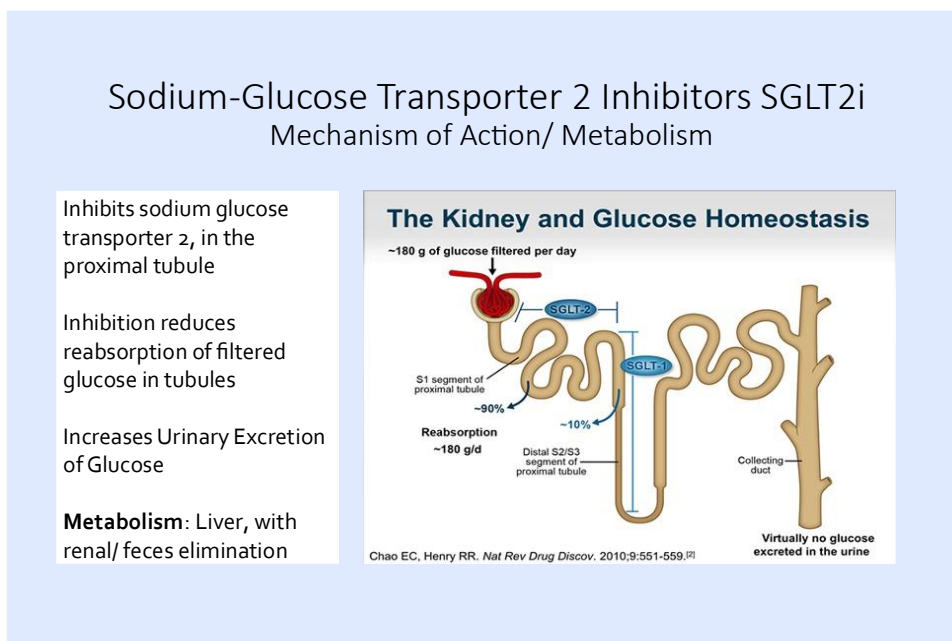
So we started with lifestyle, metformin....

- a. If ASCVD, need an agent with high efficacy for glucose lowering, or high efficacy for weight management goals, next step may be a GLP1RA (consider cost & patient preferences)

- b. If ASCVD, CHF or CKD, need to minimize hypoglycemia
 - i. Next step might have been the **SGLT2-inhibitor class (SGLT2i)**...

K. SGLT2 Inhibitors (SGLT2i) (“flozins”)

1. 1886, German physician Joseph von Mering demonstrated that the ingestion of high doses of phlorizin (from the bark of apple trees) caused people to have glucosuria
2. Modest weight loss benefit
3. No risk of hypoglycemia
4. Cardiovascular and renal benefit in RCT's
5. Can be expensive
6. Once daily pills
 - Canagliflozin
 - Dapagliflozin
 - Empagliflozin
 - Ertugliflozin



7. **Glycemic efficacy:** Mean reductions in A1C 0.4 to 1.1%
8. **Weight loss-** SGLT2 inhibitors decrease weight. In 12-week trials weight loss of 2 to 3 kg was reported.

9. **Cardiovascular effects** - associated with decreased cardiovascular events (MACE), decreased risk for heart failure and heart failure hospitalization, lower blood pressure 3-5 mmHg.
10. **Renal Effects**- These agents are associated with decrease progression of kidney disease. These agents are used for their renal benefit. Many are now ok to use down to a GFR 30 mL/min/1.73 m). But each one has specific parameters (more data coming out- fast moving area of research)
11. **Side Effects:**
 - a. **Genitourinary tract** – Most common. There is an approximate 2-4x increased incidence of vulvovaginal candidiasis, urinary tract infections.
 - b. **Necrotizing Fasciitis** of the perineum (fourniers gangrene)
 - c. **Hypotension/ volume depletion** – SGLT2 inhibitors cause an osmotic diuresis and intravascular volume contraction.
 - d. **Acute kidney injury** – some requiring hospitalization and dialysis.
 - e. **Diabetic ketoacidosis** – "Euglycemic" (usually meaning plasma glucose <250 mg/dL) diabetic ketoacidosis has been reported in patients with type 2 diabetes taking SGLT2 inhibitors.
12. SGLT2 inhibitors should not be used in patients with
 - a. Type 1 diabetes
 - b. Ketosis prone diabetes
 - c. Patients on Dialysis
13. These agents should be held for surgery, NPO states and dehydration to reduce the risk of euglycemic DKA
14. These agents are used for their renal benefit
 - a. They each have GFR recommendations (many are now ok to use down to an eGFR 30 mL/min/1.73 m)
 - b. But each one has specific parameters (more data coming out- fast moving area of research)

L. Management Strategy

Glucagon-like peptide-1 (GLP-1) receptor agonists and sodium-glucose co-transporter 2 (SGLT2) inhibitors can be initiated independent of A1C and metformin therapy, particularly for the individuals at high risk for heart failure, CKD (chronic kidney disease), and/or ASCVD (atherosclerotic cardiovascular disease). Both classes offer cardiovascular benefits and lower all-cause mortality while offering glycemic control and weight loss

M. Selecting Therapies

1. We did lifestyle changes, metformin and went over the glycemic and non-glycemic benefits of 2 classes, the GLP1RA and SGLT2i.
2. Next, agents that are LOW cost, and then some less commonly used diabetic agents

N. Sulfonylureas (SU)

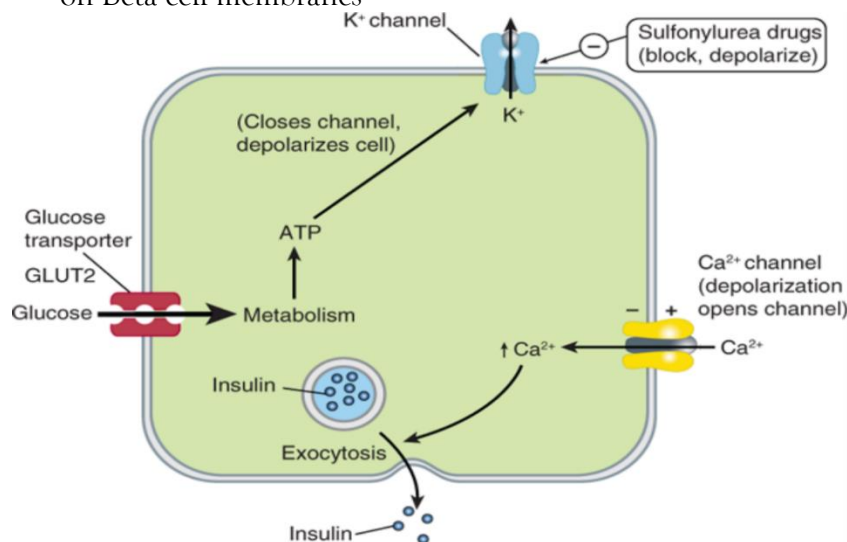
1. Enhances insulin secretion from the beta cell
2. Known for their risk of hypoglycemia
3. Associated with weight gain
4. Low cost
5. No known CVD benefits
6. Usually lower A1c approx. 1-2%

Glyburide

Glipizide

Glimepiride

7. **Mechanism of Action:** Enhances Insulin secretion by closing K-ATP channels on Beta cell membranes



Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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8. **Metabolism:** by the liver to weakly active or inactive metabolites, which are

excreted by the kidney and partly excreted in the bile

- a. Glyburide is metabolized into products with lower hypoglycemic activity
- b. Glipizide and glimepiride have inactive metabolites

9. Side effects:

- a. **Hypoglycemia** is the most common side effect. Impaired renal or hepatic function, exercise, missed meals, ethanol ingestion or use with other glucose lowering drugs are being utilized
- b. **Weight gain**
- c. **Less common**: dizziness, headache, skin reactions (rash, pruritis)
- d. **Sulfonamide allergy**: chemical similarities are present between sulfonamides and sulfonylureas. Use of sulfonylureas is not specifically contraindicated in patients with sulfonamide allergy, however a risk of a cross-reaction exists. Do not use if a previous severe sulfonamide reaction
- e. **Hypoglycemia Review**: Symptoms include sweating, heart palpitations, hunger, confusion; severe hypoglycemia can cause coma or death. Patients with frequent hypoglycemia may lose warning symptoms Treatment is simply carbohydrate (juice glucose tablets); not fats that would delay the absorption of glucose. Glucagon injection can be used in someone who cannot swallow safely.

O. Meglitinides

- 1. Similar to sulfonylurea with mechanism of action and efficacy, but rapid onset and short duration of action. Administered with meals to reduce postprandial hyperglycemia
- 2. Can cause hypoglycemia
- 3. More expensive than sulfonylureas, can be used if allergy to sulfonylurea medications
- 4. No CVD benefit
- 5. These agents are not as commonly used

Repaglinide

Nateglinide

6. **Mechanism of Action:** regulate adenosine triphosphate (ATP)-sensitive potassium channels (K-ATP channels) in beta cells to increase insulin secretion. Structurally different than sulfonylureas, exert their effects via different pancreatic beta cell receptors
7. **Metabolism**
 - a. Repaglinide metabolized by the liver, <10 percent renally excreted. Can be used safely in patients with chronic kidney disease.
 - b. Nateglinide is hepatically metabolized, with renal excretion of active metabolites. Caution if chronic liver or renal disease.

P. Thiazolidinediones (TZD's, "glitazones")

1. Decrease insulin resistance, "insulin sensitizers"
2. May be best known for side effects: fluid retention, possible precipitation or worsening of congestive heart failure and a meta-analysis showing possible increased CVD
3. Associated with weight gain
4. Inexpensive
5. Popular from late 1990's- 2007, not commonly used now due to their side effect profile and since many other newer classes of medications became available

Pioglitazone

Rosiglitazone

6. **Mechanism of action:** Insulin Sensitizers
 - a. Ligands of peroxisome proliferator-activated receptor gamma (PPAR- γ) in muscle, fat and liver, part of the steroid and thyroid superfamily of nuclear receptors
 - b. Modulates the expression of the genes involved in lipid and glucose metabolism, insulin signal transduction, and adipocyte and other tissue differentiation

- c. May increase expression of genes involved in promoting lipid storage and decrease expression of genes associated with inflammation, such as interleukin (IL)-6
- d. **Efficacy:** lowers A1C levels by 0.5 – 1.4%.
- e. **Potential benefit- MASLD or MASH**
- f. **Cardiovascular Effects:**
 - i. Both agents increase the risk of heart failure.
 - ii. May have anti-inflammatory, antithrombotic and anti-atherogenic properties, and have been shown to improve a number of cardiovascular risk factors. Pioglitazone has been associated with lower risk of myocardial infarction. Effects may depend upon the specificity of PPAR activation. Since pioglitazone effects both PPAR γ and PPAR α receptors, it has a more favorable effect on lipids.
- g. **Metabolism:** by the liver. Biologic half – life is much longer than pharmacologic half – life (can take weeks to work and weeks for their effects to resolve).
- h. **Other Side Effects:**
 - i. **Weight gain**
 - ii. **Skeletal fractures** – increasing body of evidence suggesting that TZD's decrease bone density and increase fracture risk, particularly in women.
 - iii. **Bladder cancer** concern that pioglitazone increases the risk of bladder cancer.
 - iv. **Hepatotoxicity** - severe hepatic toxicity has occasionally been reported

DPP-4 inhibitors (“gliptins”)

1. Well tolerated pills, but weaker A1c lowering (0.4-0.9%)
2. Do not cause hypoglycemia
3. Weight neutral
4. No evidence of CVD benefit

5. Can be expensive

Sitagliptin

Linagliptin

Saxagliptin

Alogliptin

6. **Mechanism of action:** Endogenous incretins, GLP-1 and GIP have a half-life only 1-2 minutes, Dipeptidyl peptidase 4 (DPP4) is the enzyme that inactivates GLP 1 & GIP

Inhibition of DPP 4 results in an increase in the amount of endogenous active GLP1, GIP

- a. GIP= Glucose-dependent insulintropic peptide. It is secreted by intestinal K cells in response to nutrient ingestion. GIP helps to stimulate glucose dependent insulin secretion for the beta cells)
7. **Metabolism:** Most are excreted unchanged renally. One (linagliptin) undergoes hepatic metabolism with biliary/ fecal excretion
8. **Side effects:** are minimal with this class, usually well tolerated
- a. Commonly reported side effects include headache, nasopharyngitis, and upper respiratory tract infection.
- b. In post marketing reports, rare hypersensitivity reactions, including anaphylaxis, angioedema, and blistering skin conditions, including Stevens-Johnson syndrome.
- c. Acute pancreatitis has been reported in association with DPP-4 inhibitors, they should not be initiated in a patient with a history of pancreatitis.

Q. Dual GLP1RA and GIPRA- the “Twincretin” Class

1. Tirzepatide (FDA approval May 2022), Injected weekly
- a. Up to 2.58 % A1c lowering (with use of highest dose)
- b. Weight reduction up to approx. 20%
- c. Not associated with hypoglycemia
- d. ASCVD data- noninferior to dulaglutide
2. **Mechanism of Action:** selectively binds to and activates both the GIP and GLP-1 receptors, the targets for native GIP and GLP-1

R. Rarely Used Agents

1. Amylin analog therapy – Pramlintide

- a. Amylin is a peptide co-secreted with insulin from the beta cells.
Pramlintide is an analog of amylin that has the same pharmacologic effects as amylin
- b. **Mechanism of action:** somewhat similar to GLP1RA
 - i. inhibits glucagon secretion, reducing hepatic glucose output
 - ii. slows gastric emptying
 - iii. works in the CNS to promote satiety
- c. **Side Effects:** Nausea (20% of patients)
- d. **Clinical Use and Efficacy:** Use in insulin treated type 2 diabetes, subcutaneous injections before meals. Lowers A1C 0.5-0.6%.
 - i. Mild reductions in weight (0.5 – 1.5kg)
 - ii. Majority of patients now use GLP1RA (better efficacy and easier to use)

2. Alpha Glucosidase Inhibitors

- a. Mechanism: slows the absorption of glucose. They inhibit intestinal alpha glucosidase (Reduces rate of polysaccharide to monosaccharide digestion in proximal small intestine)
- b. Not often used due to Side Effects: diarrhea, bloating, flatulence
 - i. (delay in the absorption of carbohydrates causes more complex carbohydrates to appear in the colon where enteric bacteria break them down generating gaseous by-products)
- c. Clinical Use and Efficacy
 - i. Only lower A1C by about 0.5%.
 - ii. Dosing is three times daily with the start of each main meal.

Acarbose

Miglitol

3. Colesevelam

- a. This agent is a bile acid sequestrant originally approved by the FDA to lower serum cholesterol levels, approved in 2008 for the treatment of type 2 diabetes.
- b. The mechanism by which it lowers glucose is unknown.
- c. Efficacy: lowers A1C by only about 0.5%
- d. Major side effect is gastrointestinal – bloating, gas and diarrhea

4. Bromocriptine

- a. Dopamine agonist.
- b. The precise mechanism of action is not known, but it appears to work on the hypothalamic neurotransmitter activities that induce and maintain the glucose resistant state of diabetes.
- c. Efficacy: lowers A1C about 0.4 – 0.5%
- d. Major side effects are nausea, vomiting, fatigue and dizziness.

Table 9.2—Features of medications for lowering glucose in type 2 diabetes

Medication (route of administration)	Glucose-lowering efficacy ¹	Hypoglycemia risk	Weight effects ²	CV effects		Kidney effects		MASH effects	Clinical considerations and adverse effects
				Effect on MACE	Effect on HF	Progression of CKD	Dosing/use considerations ³		
Metformin (oral)	High	No	Neutral (potential for modest loss)	Potential benefit	Neutral	Neutral	Contraindicated with eGFR <30 mL/min/1.73 m²	Neutral	- GI side effects: mitigate with slow dose titration, extended-release formulations, and administration with food. - Potential for vitamin B12 deficiency: monitor and replete as appropriate.
SGLT2 inhibitors (oral)	Intermediate to high	No	Loss (intermediate)	Benefit: canagliflozin, empagliflozin	Benefit: canagliflozin, dapagliflozin, empagliflozin, ertugliflozin	Benefit: canagliflozin, dapagliflozin, empagliflozin	- See labels of individual agents for dosage considerations for kidney function - Glucose-lowering effect is minimal at eGFR <45 mL/min/1.73 m² and lower; continue for cardiovascular and kidney benefit until dialysis or transplantation	Unknown	- DKA risk in individuals with insulin deficiency (rare in T2D): discontinue, evaluate, and treat promptly if suspected; be aware of predisposing risk factors and clinical presentations (including euglycemic DKA); mitigate risk with sick-day planning; discontinue before scheduled surgery (e.g., 3–4 days), during critical illness, or during prolonged fasting - Genital mycotic infections: mitigate risk with genital hygiene and avoid use in high-risk individuals - Necrotizing fasciitis of the perineum (Fournier gangrene): rare, prompt treatment if suspected - Intravascular volume depletion: attention to volume status and blood pressure, particularly when ill or fasting; adjust other volume-contracting agents as applicable; monitor kidney function upon initiation
GLP-1 RAs (SQ; semaglutide also available in oral formulation)	High to very high	No	Loss (intermediate to very high)	Benefit: dulaglutide, liraglutide, semaglutide (SQ) Neutral: exenatide once weekly, lixisenatide	Neutral	Benefit for renal and points in CVOTs, driven by albuminuria outcomes: dulaglutide, liraglutide, semaglutide (SQ) Demonstrated benefit for progression of CKD for semaglutide (SQ)	- See labels of individual agents for dosage considerations for kidney function - No dose adjustment for dulaglutide, liraglutide, or semaglutide - Monitor kidney function when initiating or escalating doses in individuals with kidney impairment reporting severe adverse GI reactions	Potential benefit	- Thyroid C-cell tumors identified in rodents; human relevance not determined - Ileus: risk level is not well established; provide guidance on discontinuation prior to surgical procedures - Pancreatitis: acute pancreatitis has been reported, but causality has not been established. Do not initiate if at high risk for pancreatitis, and discontinue if pancreatitis is suspected - Biliary disease: evaluate for gallbladder disease if cholelithiasis or cholecystitis is suspected; avoid use in at-risk individuals - Diabetic retinopathy: close monitoring of retinopathy in those at high risk (older individuals and those with longer duration of T2D [≥10 years]) - Impact on drug absorption: orally administered drug absorption may be impaired during dose titration (including of oral contraceptives) - GI side effects: counsel on potential for GI side effects; provide guidance on dietary modifications to mitigate GI side effects (reduction in meal size, mindful eating practices [e.g., stop eating once full], decreasing intake of high-fat or spicy food); consider slower dose titration for those experiencing GI challenges. Not recommended for individuals with gastroparesis.
Dual GIP and GLP-1 RA (SQ)	Very high	No	Loss (very high)	Under investigation	Under investigation	Under investigation	- See labels of individual agents for dosage considerations for kidney function - No dose adjustment - Monitor kidney function when initiating or escalating doses in individuals with kidney impairment reporting severe adverse GI reactions	Potential benefit	
DPP-4 inhibitors (oral)	Intermediate	No	Neutral	Neutral	Neutral (potential risk: saxagliptin)	Neutral	- Dose adjustment required based on kidney function (sitagliptin, saxagliptin, alogliptin) - No dose adjustment required for linagliptin	Unknown	- Pancreatitis has been reported, but causality has not been established. Discontinue if pancreatitis is suspected - Postmarketing concerns about joint pain (consider discontinuing if debilitating and other treatment options are feasible) and bullous pemphigoid (discontinue if suspected)
Pioglitazone (oral)	High	No	Gain	Potential benefit	Increased risk	Neutral	- No dose adjustment required - Generally not recommended in kidney impairment due to potential for fluid retention	Potential benefit	- Increased risk of HF and fluid retention. Do not use in the setting of HF. - Risk of bone fractures. - Bladder cancer: do not use in individuals with active bladder cancer, and use caution in those with prior history of bladder cancer.
Sulfonylureas (2nd generation) (oral)	High	Yes	Gain	Neutral	Neutral	Neutral	- Glyburide: generally not recommended in CKD - Glipizide and glimepiride: initiate conservatively to avoid hypoglycemia	Unknown	- FDA Special Warning on increased risk of CV mortality based on studies of an older sulfonylurea (tolbutamide); glimepiride shown to be CV safe (see text) - Use with caution in individuals at risk for hypoglycemia, particularly if in combination with insulin.
Insulin (human) (SQ; regular insulin also available as inhaled formulation) Insulin (analogs) (SQ)	High to very high	Yes	Gain	Neutral	Neutral	Neutral	Lower insulin doses required with a decrease in eGFR; titrate per clinical response	Unknown	- Injection site reactions - Higher risk of hypoglycemia with human insulin (NPH or premixed formulations) vs. analogs - Risk of hypoglycemia and duration of activity increases with the severity of impaired kidney function. - Refer to device-specific instructions for insulins compatible with different delivery systems (i.e., pumps, connected insulin pens, insulin patches).

Extra-glycemic Benefits in Randomized Control Trials
(observed advantages are specific to **individual agents in the class** that have demonstrated efficacy in RCT)

ASCVD (atherosclerotic cardiovascular disease)

- GLP1RA
- SGLT2i

CHF (congestive heart failure)

- SGLT2i

CKD (chronic kidney disease)

- SGLT2i
- GLP1RA (semaglutide)

Weight

Weight loss Benefit

- Strongest: Twincretin, GLP1RA
- SGLT2i
- Metformin

Weight Gain noted

- Insulin
- Sulfonylureas
- Meglitinides
- Thiazolidenediones (TZD)

Additional Cases for Study

Case 1:

The patient is a 52-year-old with a medical history significant for hypertension, obesity (BMI = 31) and hyperlipidemia who was told that they were a “borderline” diabetic 2 years ago at their last physical exam. You order routine labs and find all are normal except for a fasting blood sugar of 160 mg/dL on two occasions and a hemoglobin A1C of 7.4%. Your best initial course of therapy is:

Case 2:

The patient is a 50-year-old with a past medical history significant for coronary artery disease status post myocardial infarction with a cardiac stent placed in the LAD, HTN, hyperlipidemia, obesity, and type 2 diabetes for 6 years. Physical exam is only remarkable for a BMI of 34 and BP is 140/88.

Laboratory evaluation reveals an A1C of 9%. The rest of the laboratory evaluation is normal. They are currently taking metformin and glyburide for diabetes. What options are available to treat the diabetes?

Case 3:

The patient is a 70-year-old with a history of coronary artery disease, congestive heart failure, chronic kidney disease, hypertension, hyperlipidemia, and glaucoma who presents to you for evaluation and treatment of diabetes. They have non-proliferative diabetic retinopathy and neuropathy. They are taking glyburide 5 mg once daily and their sugars are not well controlled (running around 200). Their labs show a GFR of 35 and a hemoglobin A1C of 8.8%. On physical exam, there is bilateral lower extremity edema. What therapeutic changes should we make?

Answers to Study Questions

These answers are not meant to be all-encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. You are starting a patient on metformin. They want to know what the most common side effect is. You tell them:

- a. **Gastrointestinal side effects**
- b. Lactic acidosis
- c. Kidney dysfunction
- d. Hypoglycemia

Answer: The most common side effect to metformin are the gastrointestinal side effects such as bloating, gas, nausea and diarrhea.

Lactic acidosis is a very rare side effect.

Metformin does not cause kidney dysfunction. However, it is cleared by the kidney and therefore cannot be used in patients with a GFR less than 30.

Metformin has very low rates of hypoglycemia.

2. A patient is started on an SGLT-2 inhibitor. They want to know the effects it will have on their metabolic profile. You tell them:

- a. **Decreased a1c, decreased weight, decreased blood pressure**
- b. Decreased a1c, increased weight, decreased blood pressure
- c. Decreased a1c, decreased weight, increased blood pressure
- d. Decreased a1c, increased weight, increased blood pressure

Answer: SGLT-2 inhibitors cause glucosuria and a natriuresis. This leads to decreased a1c, decreased weight and decreased blood pressure.

3. A patient with type 2 diabetes asks you to provide them with a medication regimen that will help them to lose weight. The following combination of medications is most likely to accomplish that goal:

- a. **Metformin, GLP-1 analog, SGLT-2 inhibitor**
- b. Metformin, sulfonylurea, bromocriptine
- c. Metformin, thiazolidinedione, colesevelam
- d. Metformin, repaglinide, pramlintide

Answer: Metformin causes a 1-2 kg weight loss. GLP-1 analogs and SGLT-2 inhibitors induce a 2-3 kg weight loss.

Answer b is incorrect because sulfonylureas are associated with weight gain and bromocriptine is largely weight neutral.

Answer c is incorrect because thiazolidinediones cause weight gain and colesevelam is weight neutral.

Answer d is incorrect because repaglinide can cause weight gain.

4. You are treating a 74 - year old type 2 diabetic patient without complications with metformin and glipizide 10mg once daily. Their fasting sugars are 90-130 mg/dL and their a1c is 6.2%. However, they get frequent hypoglycemia before lunch. The best advice to give the patient is:

- a. Hold the metformin
- b. Decrease the dose of metformin
- c. Eat a mid-morning snack
- d. **Hold the glipizide**

Answer: Hypoglycemia is not a common side effect with metformin. Therefore, holding it or decreasing the dose are incorrect answers.

Eating a mid-morning snack would help decrease pre-lunch hypoglycemia, however stopping the glipizide is a better answer for many reasons. The patient's a1c is well below goal and at their age their glucose does not need to be so tightly controlled. The frequent hypoglycemia is leading to the lower a1c. Eliminating the glipizide will correct the hypoglycemia, simplify the patient's regimen and will reduce side effects and expense.

5. You start a 35-year-old patient with type 2 diabetes on a GLP-1 analog. The medical student rotating on your service asks you to explain the metabolic effects of this class of medications. You tell them that they:

- a. Increase insulin and glucagon secretion in a glucose dependent fashion.
- b. Increase insulin secretion and decrease glucagon in a glucose dependent fashion.**
- c. They are insulin analogs that acts like insulin in the body.
- d. Increase insulin secretion and decrease glucagon in a glucose independent fashion.

GLP1RA work in a **glucose dependent** fashion to increase insulin secretion and decrease glucagon. GLP1 analog agents are analogs to the incretin hormone glucagon like peptide 1, not insulin.

Answers to Cases

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

Case 1:

The patient is a 52-year-old with a medical history significant for hypertension, obesity (BMI = 31) and hyperlipidemia who was told that they were a “borderline” diabetic 2 years ago at their last physical exam. You order routine labs and find all are normal except for a fasting blood sugar of 160 mg/dL on two occasions and a hemoglobin A1C of 7.4%. Your best initial course of therapy is:

The laboratories meet the diagnostic criteria for a diabetes diagnosis. Recommend lifestyle changes (refer for diabetes education and for dietary advising), address SDOH, and start metformin if no contraindications are present. Consider GLP1-RA to address weight management and glucose control.

Case 2:

The patient is a 50-year-old with a past medical history significant for coronary artery disease status post myocardial infarction with a cardiac stent placed in the LAD, HTN, hyperlipidemia, obesity, and type 2 diabetes for 6 years. Physical exam is only remarkable for a BMI of 34 and BP is 140/88.

Laboratory evaluation reveals an A1C of 9%. The rest of the laboratory evaluation is normal. They are currently taking metformin and glyburide for diabetes. What options are available to treat the diabetes?

The patient has established atherosclerotic cardiovascular disease. Treatment with either a GLP1-RA or an SGLT2 inhibitor with cardiovascular benefit should be considered. The GLP1RA could give additional weight loss benefit. Due to the hypoglycemia risk, consider stopping the glyburide.

Case 3:

The patient is a 70-year-old with a history of coronary artery disease, congestive heart failure, chronic kidney disease, hypertension, hyperlipidemia, and glaucoma who presents to you for evaluation and treatment of diabetes. They have non-proliferative diabetic retinopathy and neuropathy. They are taking glyburide 5 mg once daily and their sugars are not well controlled (running around 200). Their labs show a GFR of 35 and a hemoglobin A1C of 8.8%. On physical exam, there is bilateral lower extremity edema. What therapeutic changes should we make?

Glyburide is a sulfonylurea and can cause hypoglycemia. In this patient, the low GFR is considered an additional risk for hypoglycemia. Consider stopping this agent. Metformin should not be started as a new agent given the low GFR.

Due to the comorbidities of coronary disease, heart failure and kidney disease, a SGLT2inhibitor with proven cardiovascular benefit should be considered. A GLP1RA could also be considered due to the ASCVD benefit. Insulin is also an option, but it would carry the risk of hypoglycemia and weight gain.